

Project Proposal

Overview:

In competitive consumer markets, maintaining an up-to-date awareness of public sentiment is vital for strategic decision-making. The DEPI-AI-Powered Brand Feedback Analysis system is a specialized machine learning pipeline and interactive dashboard designed to scrape, clean, evaluate, and rank consumer sentiment regarding brands operating within Egypt (e.g., telecom providers, banking, fintech, and e-commerce). Utilizing data extracted from Reddit, the system addresses the lack of specialized Egyptian market sentiment analysis tools by handling mixed English-Arabic code-switching, generating reliable sentiment indices, and producing real-time competitive rankings.

Objectives

- **Automated Data Ingestion:** Establish an automated interface with Reddit to collect brand-specific posts and comments using targeted English and Arabic queries.
- **Localized Text Processing:** Implement a custom text cleaning and normalization pipeline optimized for Egypt-focused dialects, diacritics, and basic stemming.
- **Accurate Sentiment Classification:** Deploy a lightweight, high-performance machine learning classifier capable of sorting unstructured feedback into positive, negative, and neutral categories.
- **Dynamic Brand Intelligence:** Formulate a multi-factor ranking engine that scores brands based on a weighted combination of sentiment distribution, user engagement, and data volume.
- **Operational Visibility:** Deliver an interactive Streamlit UI showing MLOps pipeline health, predictive confidence, and analytical brand-to-brand comparisons.

Project Scope

In Scope

- Scraping of Reddit posts & comments via JSON/OAuth API.
- Supporting Egyptian telecoms, fintech, e-commerce, and banks.
- Classical ML models (LinearSVC, Logistic Regression).
- Custom rule-based stemmers for Arabic prefixes & English plurals.
- Interactive Streamlit Dashboard for brand comparison & MLOps UI.

Out of Scope

- Scraping of closed networks (e.g., WhatsApp, private groups).
- Complex Arabic dialect translation to standard English.
- Deploying heavy Deep Learning Transformers (e.g., LLMs/GPT-4) in low-resource local runtimes.
- Real-time financial market cost implication modeling.

Milestones and Deliverables

- **Milestone 1 (System Scoping Approved):** Functional specification and risk management plan finalized.
- **Milestone 2 (Data Ingestion Verified):** Scraping modules functional, returning valid JSON objects, and saving structured raw files under data/raw/.
- **Milestone 3 (ML Model Freeze):** LinearSVC trained to $\geq 93\%$ validation accuracy and serialized with its corresponding vectorizer inside models/.
- **Milestone 4 (Score Integration Completed):** Recommendation engine calculations configured and verified against synthetic data edge cases.
- **Milestone 5 (Production Release):** Streamlit application operating locally and deployed to the hosting environment.

Risk Assessment & Mitigation Plan				
Risk Description	Probability (1-5)	Impact (1-5)	Score (1-25)	Mitigation Strategy
Reddit API Block/Rate Limiting: Scraper operations trigger IP bans or 403 HTTP errors due to structural changes or volume thresholds.	4	4	16	Utilize formal OAuth2 access token generation via registered application credentials; introduce automated <i>time.sleep(1)</i> delays between queries; construct fallback methods to scan standard web structures if APIs decline calls.
Model Concept Drift: Sentiment models trained on historical Twitter datasets struggle to accurately parse unique internet slang and modern brand terminology.	3	4	12	Implement a continuous feedback mechanism. Save raw data for local evaluation and retrain the saved model with specialized local datasets like <i>sample_for_labeling.csv</i> to capture domain-specific terminology.
Egypt Contextual Low Recall: Scrapes return high volumes of non-Egyptian content for generic keywords (e.g., "Orange" returns fruit or color references).	4	3	12	Apply strict contextual keyword matching (<i>egypt</i> , <i>مصر</i> , <i>cairo</i> , <i>egp</i> , etc.) immediately after data retrieval. Automatically trigger global fallback searches only when filtered records are insufficient.
Language Code-Switching Complexity: Sentiment scoring fails on mixed sentences containing both Arabic scripts and English expressions.	3	3	9	Ensure the cleaning regex maintains both script types: <i>[^a-zA-Z\u0600-\u06FF\s]</i> . Utilize TF-IDF n-grams to preserve contextual associations between juxtaposed Arabic and English tokens.
Runtimes & Performance Bottlenecks: Large dataset processing inside the interactive Streamlit dashboard creates UI freezes or out-of-memory errors on deployment runtimes.	2	3	6	Leverage Streamlit's structural caching directives (<i>@st.cache_data</i> , <i>@st.cache_resource</i>) to store processed brand profiles. Store pre-calculated scores instead of running full inferences on every dashboard page refresh.

Key Performance Indicators (KPIs)

Metric Category	Target Performance Benchmark
Machine Learning Accuracy	- Core Validation Accuracy $\geq 93\%$ - Macro F1-Score for Negative Class ≥ 0.90 - Minimal False Positives on neutral text
Pipeline Performance	- Scraper Latency $< 1.5s$ per query page - Batch Inference $< 5.0s$ per 100 records - Dashboard UI Load Time $< 2.0s$
System Usability	- Zero UI failures on unmapped inputs - Local run setup completed in < 4 steps - Streamlit deployment availability $\geq 99\%$